

Table 1: File compression ratio

Archiver	Mozilla Firefox		Google Earth	
	161 files 15,684,168 bytes		115 files 23,530,652 bytes	
	Compressed size	Ratio	Compressed size	Ratio
7-Zip 4.23 (7z format)	4621135	100%	6109183	100%
WinRAR 3.50	5021556	109%	6824892	112%
CABARC 5.1	5131393	111%	7434325	122%
WinZip 10.0 beta (maximum-PPMd)	5277118	114%	8200708	134%
7-Zip 4.23 (zip format)	6222627	135%	8909446	146%
WinZip 10.0 beta (maximum-portable)	6448666	140%	9153898	150%

Source: 7-zip.org

Features

Libvirt supports storage pool types such as directory backend, logical backend, disk backend, local file system backend, network file system backend, iSCSI backend, SCSI backend, multipath backend, RBD (RADOS Block Device) backend, Sheepdog backend and Gluster backend.

7-Zip: A file compression solution

7-Zip is open source software written in C++ that is used to compress files in the 7z archive format. It was developed by Igor Pavlov and first released on July 18, 1999. A stable release has been available since November 18, 2010. It supports various operating systems such as GNU/Linux, Microsoft Windows and Mac OS X. Table 1 gives the file compression ratio of this storage solution.

It has a graphical user interface and command line interface.

Features

- Confidentiality increases due to AES-256 encryption in 7z and ZIP formats
- Supports localisation for 79 languages
- It has a powerful command line version and file manager
- Provides integration with Windows Shell

TrueCrypt: An encryption solution

TrueCrypt is an open source application used for data encryption before it is saved and data decryption before it is consumed. It supports operating systems such as Linux, OS X, Microsoft Windows, etc. TrueCrypt currently uses the XTS mode [XEX-based Tweaked CodeBook mode (TCB) with CipherText Stealing (CTS)]. Password, keyfile(s) or correct encryption keys are only ways to access data stored on an encrypted volume. They are used to protect data confidentiality even if the OS is not active.

Supported algorithms: Twofish, Serpent and AES

Supported hybrid combinations of algorithms:

Cascades, Serpent-Twofish-AES, Serpent-AES, AES-Twofish-Serpent, Twofish-Serpent and AES-Twofish

Supported cryptographic hash functions: TrueCrypt supports the SHA-512, RIPEMD-160 and Whirlpool hash algorithms. Roadmap for future versions includes features such as support for Windows 8 to encrypt Windows system partitions/drives on UEFI-based computers, 'raw' CD/DVD volumes, command line options for volume creation on the Windows operating system, etc. You can download these from: <http://www.truecrypt.org/downloads>

Features

- Two-factor authentication
- Individual disk partitions and swap space can be encrypted
- Encrypted container can be persisted in a file
- Hidden containers - one per 'outer' container is supported; hidden containers are similar to nested containers
- Pre-boot authentication on Windows is required before booting the computer to encrypt the boot disk
- More than one active key
- Passphrase strengthening - key strengthening is used with pure text passwords
- Hardware acceleration - cryptographic accelerator expansion cards can be used
- Windows MBR volumes can be encrypted and not UEFI GPT drives
- Complete physical disk or logical volume, including the partition tables and master boot record can be encrypted 

References

- <http://www.truecrypt.org>
- <http://en.wikipedia.org/wiki/FreeNAS>
- http://doc.freenas.org/index.php/Quick_Start_Guide

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